

Case: MW Petroleum Corporation (A)

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I. Executive Summary

Apache Corporation considers the acquisition of MW Petroleum with the intention of expanding its reserve portfolio and enhancing long-term production capacity. The acquisition of MW Petroleum would increase Apache's proved developed and undeveloped reserves, while providing upside to their probable and possible reserves. Given the volatility of oil prices, the value of acquiring MW Petroleum extends beyond static cash flows and includes the options in the timing of reserve development.

Using a discounted cash flow (DCF) approach with a 14.18% risk-adjusted discount rate, we estimate the total DCF value of MW Petroleum at approximately \$483.7 million (Exhibit B). The total DCF value of MW Petroleum reflects the present value of proved developed, proved undeveloped, probable, and possible reserves based on the projection of operating cash flows and terminal value assumptions. However, MW Petroleum's value contains embedded real options associated with the decision to delay or expand the development of undeveloped reserves beyond its DCF estimated value. When incorporating the Black-Scholes framework using a present value of underlying cash flows (S) of \$64.42 million, a development cost of \$31.37 million, a risk-free rate of 8.03%, time to expiration of seven years, and oil price volatility of 40.00%, we estimate the option value to be approximately \$48.33 million (Exhibit E). The option value of developing the undeveloped reserves exceeds the static net present value of \$33.05 million, indicating that managerial flexibility adds incremental value beyond the traditional DCF.

After accounting for the value of real options, the adjusted economic value of MW Petroleum increases above the base DCF estimate. Under these assumptions, the acquisition appears economically justified, demonstrating that Apache is able to capture both the projected operating cash flows and the value of the embedded flexibility. This allows us to conclude that having the ability to exercise real options enhances the attractiveness of the transition in a volatile commodity environment.

II. Data and Key Assumptions

The valuation of MW Petroleum is conducted using an adjusted present value (APV) framework, which separates the value of operating assets from the value created by financing. The unlevered cost of capital is derived using CAPM. With an unlevered (asset) beta of 0.82, a market risk premium of 7.5%, and a risk-free rate of 8.03%, the resulting unlevered cost of capital is approximately 14.18%. This rate is used to discount the operating cash flows of all reserve categories. The expected return on debt post-acquisition is assumed to be 12%, and the corporate tax rate is 35.7%. Under the borrowing constraint, debt capacity is limited to 50% of the value of proved developed and proved undeveloped reserves. Probable and possible reserves provide no collateral value and therefore have zero debt capacity.

Projected annual cash flows for each reserve category are summarized in Exhibit A. These projections include operating cash flows and terminal values for proved developed, proved undeveloped, probable, and possible reserves. Exhibit B reports the discounted cash flow valuation of each category using the 14.18% unlevered cost of capital. Exhibit C then applies the 50% debt-to-value constraint to the proved reserve categories each year, generating a declining debt capacity profile as reserves are depleted. Annual interest expense is calculated as 12% of the allowable debt level, and the associated tax shield equals interest multiplied by the corporate tax rate. The present value of these tax shields is obtained by discounting them at the cost of debt. This structure ensures that operating value (Exhibit B) and financing benefits (Exhibit C) are treated separately and consistently within the APV framework.

III. Discounted Cash Flow Valuation

The DCF valuation process begins with the projected cash flows in Exhibit A. For proved developed reserves, the annual operating cash flows decline gradually over time as production decreases. These projected cash flows, along with the terminal value, are discounted at the unlevered cost of capital to determine the present value of proved developed reserves, as reported in Exhibit B. Because these reserves are already in production and do not require discretionary development decisions, their value is derived directly from expected operating performance.

Proved undeveloped reserves require development expenditures before production begins. Exhibit A shows early negative or smaller cash flows reflecting development costs followed by positive operating cash flows in later years. The DCF valuation discounts the entire projected stream, including terminal value, at the unlevered cost of capital to obtain their present value in Exhibit B. This base-case valuation assumes immediate development for purposes of calculating operating value.

Probable and possible reserves are valued similarly by discounting their projected operating cash flows and terminal values at the same unlevered cost of capital. As shown in Exhibit B, these categories contribute additional operating value to the firm. However, unlike proved reserves, they do not contribute to borrowing capacity under the 50% constraint.

The debt schedule in Exhibit C applies the borrowing base rule each year by setting debt capacity equal to 50% of the value of proved developed and proved undeveloped reserves. As those reserve values decline, allowable debt decreases accordingly. Interest expense is calculated annually on the permitted debt level, and the resulting tax shield is discounted at the cost of debt to determine its present value. The overall APV is therefore calculated as the sum of the unlevered operating value of all reserve categories (Exhibit B) and the present value of tax shields (Exhibit C).

IV. Real Options Analysis

MW Petroleum contains several embedded real options arising from managerial flexibility in developing its reserves. The most important are the option to delay development of proved undeveloped reserves, the option to invest in proving and developing probable reserves, and the option to develop possible reserves. Each of these resembles a call option: management can invest if oil and gas prices are favorable and refrain from investing if conditions deteriorate. Proved undeveloped reserves reflect a timing option, probable reserves involve additional engineering discretion, and possible reserves function like deep out-of-the-money calls given their higher uncertainty and required capital.

Beyond these development-related options, MW also benefits from broader timing flexibility across its investment program, an implicit abandonment option if prices fall sharply, and the potential for operational improvement under Apache's management. However, only the first three options were quantified because they are directly tied to clearly defined reserve categories with identifiable investment costs and deferral periods, making them suitable for option-style modeling. The remaining options are more diffuse or interact with other flexibilities and valuing them separately would require additional assumptions and risk double counting. As a result, they are discussed qualitatively, while the reserve-specific options are incorporated into the valuation.

We modeled the option to delay development of each reserve category using the Black-Scholes framework. The development of proved undeveloped, probable, and possible reserves requires substantial and mostly

irreversible capital expenditures, creating a real option of deferring the investment and waiting for more favorable market conditions.

For the proved undeveloped reserved, we identified the aggressive capital expenditures spike in years 1 and 2. To isolate the underlying asset value (S), we added back these development expenditures to the free cash flows and recalculated the discounted cash flow value. The strike price (K) was defined as the present value of the development expenditures in years 1 and 2, discounted at the risk-free rate of 8.03% (Exhibit D.1). Using a baseline volatility of 40% (based on historical oil and gas price volatility), a risk-free rate of 8.03%, and a deferral period of 7 years, we computed the option value of \$48.33 million (Exhibit E).

Applying the same structure to the probable and possible reserves, we computed option values of \$49.17 million and \$37.57 million, respectively (Exhibit E). Note that for these segments, capital expenditures are concentrated in the first five years, so the strike price was defined as the present value of capital expenditures over years 1–5 (Exhibits D.2 & D.3). The computed call option values indicate that a significant portion of MW's value is driven by the flexibility to delay investment.

Based on the calculated call option values, Apache should delay exercising each development option as long as possible (up to the 7-year limit). Given that there is no penalty from deferral, waiting preserves upside potential and limits downside risk. This is further supported by our sensitivity analysis. As the time to expiration (T) increases, option values rise across all volatility assumptions. Similarly, higher volatility increases option value, especially for the probable and possible reserves (Exhibits F1, F.2, & F.3). As per the sensitivity analysis, the price volatility is high, reinforcing the value of waiting. Based on a comparison of segment static NPVs and their respective call option values, possible reserves should be exercised last. While their static NPV is near zero, their call value is \$37.57 million due to increased uncertainty and associated irreversible investment. Furthermore, proved undeveloped reserves, while less uncertain, still benefit from delayed exercise unless commodity prices are immediately favorable. Consequently, Apache should treat the develop the reserves sequentially as market conditions justify investment.

V. Conclusion and Strategic Implications

We valued MW Petroleum to be approximately \$483.7 million based solely on our discounted cash flow analysis and using a 14.18% unlevered cost of capital. Once we incorporated the additional value from embedded real options using the Black-Scholes framework, particularly the option to delay development of proved undeveloped reserves, the total economic value increased by approximately \$48.33 million. The option-adjusted valuation then exceeded the static DCF estimate, reflecting the importance of managerial flexibility in a volatile commodity environment. If the acquisition price is below or near this adjusted valuation, MW Petroleum will be economically attractive for Apache Corporation.

With respect to financing, the leverage constraint does not materially bind under the assumptions provided. Apache retains sufficient borrowing capacity to complete the transaction without breaching their financial limits. While incremental leverage does increase financial risk, it does not eliminate the positive net present value of the acquisition under our base-case assumptions. The real options embedded in MW's Petroleum undeveloped reserves increase their value relative to relying solely on a traditional DCF approach. While the static NPV of the proved undeveloped reserves is \$33.05 million, the Black-Scholes valuation produces an option value of approximately \$48.33 million under an implied oil price volatility of 40.00% and a seven-year window for development. The difference between the values illustrates that flexibility to defer development enhances value in uncertain markets.

Finally, Apache should consider delaying the development of undeveloped reserves if oil prices are weak or volatile. The option to wait allows the firm to preserve capital and invest only when price conditions are

favorable. Given the high volatility of oil prices, deferring development until market signals improve may maximize long-run value. MW Petroleum represents a strategically valuable acquisition opportunity, particularly when accounting for embedded real options and the flexibility to optimally time development decisions.

VI. EXHIBITS

Exhibit A: Reserve Categories and Cash Flow Summary

	1991	1992	1993	1994	1995	1996	1997	1998
Cash Flows	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8
Proved Developed	89.40	80.00	69.90	63.00	56.00	49.50	44.90	41.90
Proved Undeveloped	(13.50)	(5.80)	7.60	5.20	4.60	6.70	9.70	12.50
Probable	(5.80)	3.80	(1.50)	(2.50)	9.70	11.90	13.10	13.00
Possible	(8.60)	(2.90)	(13.50)	(28.40)	(17.10)	2.50	12.50	13.40

	1999	2000	2001	2002	2003	2004	2005	Terminal
Cash Flows	Year 9	Year 10	Year 11	Year 12	Year 13	Year 14	Year 15	Year
Proved Developed	37.60	34.30	34.20	29.50	26.10	23.50	22.10	92.10
Proved Undeveloped	16.00	15.10	6.80	14.70	15.20	16.50	15.10	67.80
Probable	13.20	11.00	10.20	10.00	8.40	7.00	6.40	51.00
Possible	18.50	16.10	21.20	20.10	18.80	16.80	14.40	72.30

Exhibit A summarizes MW Petroleum's projected cash flows by reserve category, including proved developed, proved undeveloped, probable, and possible reserves. These projections are based on the basis for the discounted cash flow valuation and the identification of embedded real options.

Exhibit B: Discounted Cash Flow Valuation of MW Petroleum

	Proved Developed	Proved Undeveloped	Probable	Possible	Total
PV of Cash Flows	352.75	26.24	30.20	(6.78)	402.40
PV of Terminal Value	12.60	9.28	6.98	9.89	38.75
Total PV by reserve type	365.35	35.51	37.18	3.11	483.70

Exhibit B presents the DCF valuation of MW Petroleum's reserves using a 14.18% risk-adjusted discount rate. It includes the present value of proved developed, proved undeveloped, probable reserves, and possible reserves.

Exhibit C: Debt Capacity and Capital Structure

	1991	1992	1993	1994	1995	1996	1997
	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7
Debt Capacity	190.90	180.87	167.92	157.63	149.68	142.81	135.76
Interest	22.91	21.70	20.15	18.92	17.96	17.14	16.29
Tax Shield	8.18	7.75	7.19	6.75	6.41	6.12	5.82
PV of Tax Shield	7.30	6.18	5.12	4.29	3.64	3.10	2.63

	1998	1999	2000	2001	2002	2003	2004	2005
	Year 8	Year 9	Year 10	Year 11	Year 12	Year 13	Year 14	Year 15
Debt Capacity	127.81	119.13	111.33	106.61	99.63	93.11	86.31	79.95
Interest	15.34	14.30	13.36	12.79	11.96	11.17	10.36	9.59
Tax Shield	5.47	5.10	4.77	4.57	4.27	3.99	3.70	3.42

PV of Tax Shield	2.21	1.84	1.54	1.31	1.10	0.91	0.76	0.63
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This exhibit presents MW Petroleum's annual debt capacity under the 50% debt-to-value constraint applied to proved developed and proved undeveloped reserves. Debt capacity is calculated as 50% of the present value of proved reserves each year. The exhibit reports the corresponding annual interest expense, the resulting tax shield (calculated as interest multiplied by the corporate tax rate), and the present value of the tax shield discounted at the project's cost of debt. The declining debt capacity over time reflects reserve depletion and the associated reduction in collateral value. This exhibit supports the evaluation of whether the leverage constraint binds and quantifies the value of financing benefits embedded in the transaction.

Exhibit D.1: Present Value of Proved Undeveloped Reserves (\$ Calculation)

	1991	1992	1993	1994	1995	1996	1997	1998
	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8
FCF	4.00	11.90	7.60	5.20	4.60	6.70	9.70	12.50
PV of FCF	3.50	9.13	5.11	3.06	2.37	3.02	3.83	4.33

	1999	2000	2001	2002	2003	2004	2005	Terminal
	Year 9	Year 10	Year 11	Year 12	Year 13	Year 14	Year 15	Year
FCF	16.00	15.10	6.80	14.70	15.20	16.50	15.10	67.80
PV of FCF	4.85	4.01	1.58	2.99	2.71	2.58	2.07	9.28
Value								64.42

Exhibit D.1 calculates the present value of operating cash flows from proved undeveloped reserves, excluding development costs. The resulting value of 64.42 represents the underlying asset (\$S) used in the Black-Scholes real option valuation.

Exhibit D.2: Present Value of Probable Reserves (\$ Calculation)

	1991	1992	1993	1994	1995	1996	1997	1998
	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8
FCF	4.20	8.10	9.90	11.50	12.30	11.90	13.10	13.00
PV of FCF	3.68	6.21	6.65	6.77	6.34	5.37	5.18	4.50

	1999	2000	2001	2002	2003	2004	2005	Terminal
	Year 9	Year 10	Year 11	Year 12	Year 13	Year 14	Year 15	Year
FCF	13.20	11.00	10.20	10.00	8.40	7.00	6.40	51.00
PV of FCF	4.00	2.92	2.37	2.04	1.50	1.09	0.88	6.98
Value								66.47

Exhibit D.2 calculates the present value of operating cash flows from probable reserves, excluding development costs. The resulting value of 66.47 represents the underlying asset (\$S) used in the Black-Scholes real option valuation.

Exhibit D.3: Present Value of Possible Reserves (\$ Calculation)

	1991	1992	1993	1994	1995	1996	1997	1998
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	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8
FCF	1.10	6.90	8.90	10.50	10.30	2.50	12.50	13.40
PV of FCF	0.96	5.29	5.98	6.18	5.31	1.13	4.94	4.64

	1999	2000	2001	2002	2003	2004	2005	Terminal
	Year 9	Year 10	Year 11	Year 12	Year 13	Year 14	Year 15	Year
FCF	18.50	16.10	21.20	20.10	18.80	16.80	14.40	72.30
PV of FCF	5.61	4.27	4.93	4.09	3.35	2.62	1.97	9.89
Value								71.17

Exhibit D.3 calculates the present value of operating cash flows from possible reserves, excluding development costs. The resulting value of 71.17 represents the underlying asset (S) used in the Black-Scholes real option valuation.

Exhibit E: Black-Scholes Real Option Valuation

	Proved Developed	Probable	Possible
Asset Value (S)	\$64.42	\$66.47	\$71.17
Development cost (K)	\$31.37	\$34.03	\$82.33
Risk-free rate (r)	8.03%	8.03%	8.03%
Oil price volatility (σ)	40.00%	40%	40%
Time to expiration (T)	7	7	7
d1	1.74	1.69	0.92
d2	0.68	0.63	-0.14
N(d1)	0.96	0.95	0.82
N(d2)	0.75	0.74	0.45
Call option value	\$48.33	\$49.17	\$37.57

Exhibit E applies the Black-Scholes model to value the option to delay development of proved undeveloped, probable, and possible reserves. Inputs include the present value of underlying cash flows (S), development cost (K), risk-free rate, oil price volatility, and time to expiration. The resulting option value is compared to the static NPV.

Exhibit F.1: Sensitivity Analysis of Real Option Value for Proved Undeveloped

Years to Wait / Volatility	30%	35%	40%	45%	50%
1	\$ 35.49	\$ 35.53	\$ 35.62	\$ 35.76	\$ 35.96
2	\$ 37.83	\$ 38.01	\$ 38.29	\$ 38.67	\$ 39.14
3	\$ 40.02	\$ 40.33	\$ 40.77	\$ 41.31	\$ 41.95
4	\$ 42.04	\$ 42.44	\$ 42.99	\$ 43.64	\$ 44.39
5	\$ 43.89	\$ 44.35	\$ 44.97	\$ 45.70	\$ 46.51
6	\$ 45.57	\$ 46.08	\$ 46.74	\$ 47.51	\$ 48.37
7	\$ 47.11	\$ 47.65	\$ 48.33	\$ 49.13	\$ 50.00

Exhibit F.1 presents sensitivity of the development option value to changes in oil price volatility and time to expiration. It demonstrates how uncertainty and managerial flexibility affect MW Petroleum's economic value.

Exhibit F.2: Sensitivity Analysis of Real Option Value for Probable

Years to Wait / Volatility	30%	35%	40%	45%	50%
1	\$ 35.09	\$ 35.16	\$ 35.28	\$ 35.47	\$ 35.72
2	\$ 37.66	\$ 37.90	\$ 38.25	\$ 38.71	\$ 39.25
3	\$ 40.07	\$ 40.45	\$ 40.96	\$ 41.60	\$ 42.32

4	\$ 42.27	\$ 42.75	\$ 43.38	\$ 44.13	\$ 44.96
5	\$ 44.28	\$ 44.83	\$ 45.53	\$ 46.35	\$ 47.26
6	\$ 46.11	\$ 46.70	\$ 47.45	\$ 48.31	\$ 49.26
7	\$ 47.77	\$ 48.39	\$ 49.17	\$ 50.05	\$ 51.01

Exhibit F.2 presents sensitivity of the development option value to changes in oil price volatility and time to expiration. It demonstrates how uncertainty and managerial flexibility affect MW Petroleum's economic value.

Exhibit F.3: Sensitivity Analysis of Real Option Value for Possible

Years to Wait / Volatility	30%	35%	40%	45%	50%
1	\$ 6.58	\$ 8.00	\$ 9.42	\$ 10.83	\$ 12.25
2	\$ 12.41	\$ 14.35	\$ 16.27	\$ 18.17	\$ 20.06
3	\$ 17.39	\$ 19.61	\$ 21.81	\$ 23.99	\$ 26.12
4	\$ 21.80	\$ 24.18	\$ 26.54	\$ 28.86	\$ 31.15
5	\$ 25.76	\$ 28.22	\$ 30.66	\$ 33.07	\$ 35.42
6	\$ 29.35	\$ 31.84	\$ 34.31	\$ 36.75	\$ 39.13
7	\$ 32.63	\$ 35.10	\$ 37.57	\$ 40.00	\$ 42.37

Exhibit F.3 presents sensitivity of the development option value to changes in oil price volatility and time to expiration. It demonstrates how uncertainty and managerial flexibility affect MW Petroleum's economic value.